Operator assistance voice automation trials

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BT's operator assistance centres, contacted by dialling 100, handle a wide variety of requests from customers for help in using the telephone service. Speech technology has now reached a maturity where it can be used to partially automate some operator functions without affecting the quality of service received by customers. Operator efficiency is improved, which not only reduces the cost of providing the operator assistance service but also frees operators from having to spend time handling the routine, repetitive calls, allowing them to concentrate on the calls that really need their attention. This paper describes a series of trials that explore the potential cost savings gained in automating some operator assistance services using BT's interactive speech applications platform.

1. Introduction

BT's operator assistance centres receive 108 million calls per year from customers requiring assistance with making telephone calls, enquiries of a general nature or requests for a wide range of telephony services such as alarm calls, reverse charge calls, advice of duration and charge. A further 19 million calls are received from members of the public wishing to be connected to one of the emergency authorities.

Currently there are 15 full-time centres handling these calls for mainland Britain. Each centre employs up to 50 operators at peak times to ensure that calls are dealt with promptly, efficiently and courteously.

This paper describes the results of trials run from April 1993 to July 1994 which automated some of the operator assistance services using the latest speech technology. The benefits to the business of reducing operator call-handling times are presented, as is a measurement of the customer's acceptance of the automated services. The technical solution adopted for the trials is also described.

The trials were held at the Leicester operator assistance centre in the Birmingham sector. This centre was chosen because it had an average operator call-handling time and, as such, any reduction of call-handling times would represent typical savings for the whole of the country.

2. Candidates for automation

A detailed data collection exercise at Leicester identified five suitable services for automation — referrals and onward connects, changed number interception, alarm call delivery advice of duration and charge, and reverse charge call delivery.

2.1 Referrals and onward connects

When a customer contacts the operator by dialling 100 and requests a service that is not provided by operator assistance (e.g. directory enquiries, residential faults) normally the operator will tell the customer the correct number to contact and release the call (referral). Only if the customer is experiencing difficulties will the operator connect the call (onward connect). Referrals and onward connects for other service numbers account for approximately 10% of the total operator assistance traffic. The most popular referral is for national directory assistance with over 50% of referral calls. This is followed by residential sales (15%) and residential faults (10%). Referrals to timeline, international operator and directory assistance, business sales and faults, and telemessages, account for the remaining 20% of traffic. With an operator taking on average about 50 seconds to complete each call, referrals and onward connects are an obvious candidate for automation.

2.2 Changed number interception

If a customer's telephone number is temporarily or permanently changed, they may request that all calls to the original number are intercepted by an operator, so callers can be informed of the new number to dial. Although this service only accounts for less than 0.1% of the total operator traffic, it can be fully automated with no operator involvement and can also offer call completion. The average call-handling time for an operator is approximately 40 seconds.

2.3 Alarm call delivery

The alarm call service is split into two stages. Firstly, a customer dials 100 to request an alarm call; the operator records the telephone number to be called and books the alarm request to mature at the required time and date. The second stage of the service is when the alarm request matures and it is presented to an operator who will call the number and wait for the call to be answered. This is the stage that can be automated. If there is no answer, or the line is busy, the alarm request is rebooked to mature ten minutes later, when an operator will attempt to deliver the alarm call again. On average, each matured alarm call takes 50 seconds from call set-up to answer. Although alarm call deliveries account for only 1.4% of traffic, they are concentrated in between the hours of 4.00 am and 8.00 am and so require extra operators to be on duty in the early hours of the morning. Automating the alarm call delivery stage significantly reduces the staffing levels required during this period of the day.

2.4 Advice of duration and charge

Advice of duration and charge is provided when a customer wants to know the cost of an operator-connected call. The customer dials 100 and asks for advice of duration and charge on the following call. The operator sets up the call and, when the call is finished, the charge information is presented to the next free operator, who will call the customer and inform them of the cost of the call. The delivery of advice of duration and charge calls account for 0.15% of the total traffic and take approximately 100 seconds to deliver. This category of call can be fully automated.

2.5 Reverse charge

Reverse charge calls occur when the customer calling 100 asks to be connected to a number, and for that number to be charged for the call. The operator calls the requested number and asks if they will pay for the call. If they agree to accept the charges, the operator will connect the two parties. Reverse charge calls represent over 23% of the total traffic, spread more or less evenly throughout the working day, tailing off towards midnight. A reverse charge call takes approximately 70 seconds of operator time to complete, of which 11 seconds is waiting for the called number to answer, and a further 16 seconds to conclude the call and to connect the two parties.

3. System environment

3.1 Architectural overview

The environment in which operator assistance calls are received is shown in Fig 1. The digits dialled by the customer (100) are translated by the local exchange into the service address code of the BT operator services system (BTOSS) serving that geographical area. The service address code will route the call across the public switched telephone network (PSTN) to the host local System X exchange of the BTOSS. The BTOSS will place the call in the 'ASSIST' queue of the appropriate operator assistance centre to be answered by an operator.

3.2 BT operator services system

BTOSS is a sub-system of the System X digital exchange. The equipment located at the System X exchange is in two parts — the operator processor function and operator conference bridge function. The purpose of the former is to process an incoming call and distribute it to an operator call-handling centre. This call distribution is dependent on traffic levels, the number of free operators at a centre and the configuration of the BTOSS. Each conference bridge allows two customers and an operator to speak to each other. There can be up to five call-handling centres connected to a BTOSS host.

Call-handling centres contain one or more operator distribution units which enable a number of operator workstations to be connected to the host. Each BTOSS host can have up to seven of these distribution units connected, each of which can have a maximum of twenty operator workstations connected to it.

An operator logs on to a workstation by inserting their headphone jackplug into the workstation headphone socket, and entering a user identity and password. A workstation is made ready for answering incoming calls by an operator pressing the 'READY' key on the workstation keyboard. As soon as an incoming call has been sent to the workstation by the BTOSS host, details of the call are immediately displayed on the screen. This includes information such as the telephone number of the originating caller, the

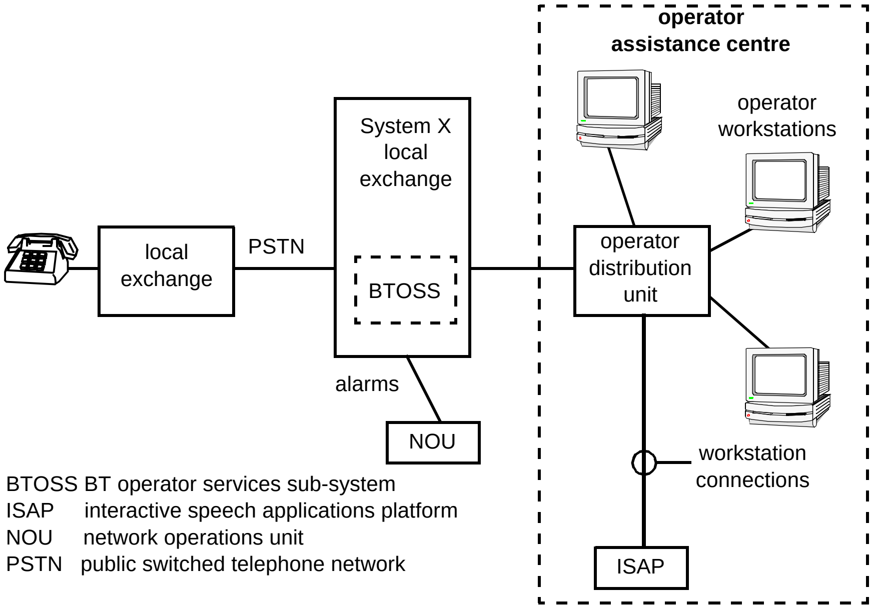

Fig 1 Operator automation trial system environment.

equipment type (e.g. business, residential, payphone), and any special facilities the customer may have, such as incoming calls barred. The operator can speak to the caller to determine the nature of the enquiry and set up a call to a third party, requeue the call to another operator or relinquish control of it. When a call is set up to a third party, the operator has full control of who speaks to whom, and can monitor a call in progress. In addition, an operator can assign various charging levels to a call related to the time and duration of the call, as well as applying a facility fee for some services such as reverse charge.

3.3 *Interactive speech applications platform*

The method chosen for implementing the trial system was based on the interactive speech applications platform (ISAP).

The central functionality of the ISAP architecture is its ability to connect to 30-channel PCM systems carrying digital speech, and run an application dialogue which can include speech recognition, speech recording, playback of pre-recorded messages, etc, as well as the capability to detect TouchTone input and call progress tones. The ISAP was designed to be both flexible and scalable. Components may be installed in various configurations to suit the needs of the application.

The ISAP is connected to an operator distribution unit as shown in Fig 1 and simulates a number of operator workstations. With the exception of changed number interception calls, all other calls that are presented to ISAP for completion will have been initially answered by an operator. Call information from the network, e.g. telephone number (calling line identity), and information obtained from the customer by the operator, e.g. reverse charge number, are contained in the screen fields of the workstation. When a call is presented to ISAP, the information in the screen fields is interpreted by ISAP to determine the type of service required. Changed number interception calls are routed directly to ISAP by BTOSS, without presentation to an operator.

An important modification to the basic ISAP software and hardware was required in order to allow the service application software to use the same call-control functions as a normal operator on the presented calls. A software component was developed which provided a high-level call control interface to the application developers. This interprets the real time screen changes and field messages that are received over the workstation interface into real time events such as incoming call received, call cleared, etc. It also translates the operator call-control requests, such as requeue, call set-up, charge and release, into the appropriate keypress and command messages that a workstation normally sends to BTOSS.

The physical interface comprises a four-wire analogue speech path and a four-wire RS422 connection for signalling. This requires the addition of intelligent serial input/output cards to the real time component of the ISAP and also a digital multiplexer unit in order to convert between the ISAP's digital speech input/output to the operator distribution unit's analogue input/output. At the time of the trials, the digital connection between the distribution unit and the BTOSS host could not be used for commercial reasons.

The application software is broken into a number of modules, each one dealing with the basic flow of each of the services described in section 2.

3.4 Dialogues

Spoken output from the ISAP to a customer is assembled from a number of pre-recorded announcement elements. These announcement elements are derived from voice recordings by a professional speaker under studio conditions. Use of a professional speaker is very important to the quality of the final application messages as they are able to provide consistent delivery, intonation and speed from the prepared scripts of announcements. The original recordings are carefully edited to ensure that the final application announcement, constructed of concatenated announcement elements with appropriate silences interspersed, will sound natural. Following the editing process, the announcement elements are digitally encoded at 64 kbit/s. Announcement elements may contain whole sentences or phrases, while others are single numbers spoken with particular intonations. For instance, each number of the code 01473 will have a different intonation depending upon its position within the sequence. In particular, special rules are applied to ensure the recommended style for operators is followed for telephone numbers, times of day, monetary amounts, time duration, cardinal and ordinal numbers.

4. Trial services

4.1 Referrals and onward connects

A customer dials 100 and asks to send a telemessage. An operator enters the abbreviation 'TGM' in the remarks field on the workstation and requeues the call which is represented on an ISAP channel. The ISAP takes control of the call and speaks out the message: *'I'm sorry, this service is not available on one hundred. Telemessage is a chargeable service and can be contacted free on one nine oh. I repeat, Telemessage is a chargeable service and can be contacted free on one nine oh.'* Following this announcement, the call is released.

A few services such as Telemessage are given a referral to inform the customer that the service is not available via operator assistance, and to encourage them to dial Telemessage directly in future. For many services, such as national directory assistance, a different announcement offers onward connection to the correct number, e.g: *'I'm sorry, this service is not available on one hundred. Directory Assistance is a chargeable service and can be contacted directly on one nine two. If you wish me to connect you please hold on, otherwise please replace the handset.'* Following this announcement, the ISAP pauses for approximately three seconds to give the customer a chance to clear down, then continues to connect the call. The correct number to dial is still spoken out first, to encourage the customer to dial direct in future. If the call originates from a payphone, a slightly different message will be spoken, as directory assistance from a payphone is free.

Other services, such as Timeline, only offer onward connect to calls that originate from a chargeable line. Calls from non-chargeable lines, e.g. payphones, are given a referral only. Calls to business and residential faults are offered onward connection for all phone types, as are calls to business and residential sales during office hours. When the offices are closed, a referral announcement is given which includes the opening times.

4.2 Changed number interception

When a customer dials a number which has been placed on operator CNI, the call is directed to an ISAP channel, which extracts the new and old number information from the call details. If the call is from a payphone the customer is given the announcement: *'The number you have dialled has changed. The new number is* <new number>. *I repeat, the new number is* <new number>. *Please redial using the new number.'* If the call is from a chargeable line, the alternative announcement is given: *'The number you have dialled has changed. The new number is* <new number>. *Please hold while I connect you.'* Before releasing the call, the ISAP checks that the line is ringing. If the number is engaged the customer is given the announcement: *'I'm sorry, the number you require is engaged at the moment. The new number is* <new number>. *Please try later.'* Other line conditions that can be encountered, such as 'number unobtainable' or 'out of service', indicate a problem and the call is requeued to an operator for manual intervention.

The part of the announcement which plays out the new number is created by concatenating discrete voice messages containing individual spoken numbers. The correct intonations and pauses are selected on the basis of the number of digits in the area code and telephone number so as to follow a set of rules which allow for maximum intelligibility and memorability (and also take into account the format that most customers prefer).

4.3 Advice of duration and charge

On completion of a call for which advice of duration and charge has been requested, the call details are presented to the ISAP. The ISAP calls the telephone number of the customer, and upon answering is played the message: *'Good morning/afternoon/evening. The cost of your call to* <number> *was* <cost> *including VAT. The duration of the call was* <time>. *Please hold if you wish to speak to an operator.'* The appropriate salutation is given depending upon the time of day and the telephone number is spoken following the same rules used for CNI above. Similarly the cost and time portions of the message are made up from segments of recordings so as to follow a set of rules for speaking monetary amounts and time duration.

4.4 Alarm call delivery

When a booked alarm call on BTOSS matures, the call details are presented to the ISAP, which calls the customer's number. When the call is answered, the ISAP plays the message: *'Good morning. This is your* <time> *alarm call. I repeat, this is your* <time> *alarm call. Please hold if you wish to speak to an operator.'* There follows a pause of approximately 8 seconds before requeuing the call back to an operator. This gives the customer time to replace the handset (particularly important if they have just been woken up). Note that the time portion of the message is the time that the alarm call is booked for — not the current time. This prevents confusion if the delivery of the call is delayed due to the customer's line being engaged.

4.5 Reverse charge

A customer dials 100 and requests a reverse charge call to a particular number. The operator types this number into the call details, adds the abbreviation 'XFC' and requeues the call to the ISAP. The ISAP immediately responds with: *'This is an automated service. Please state your name after the tone.'* There then follows a fairly lengthy beep, similar to that heard on most answering machines, after which the ISAP allows a short window of approximately three seconds to allow the caller to record their name, e.g. *'Joe from the office.'* As soon as a name is recorded, or if no response is detected within one second, the following message is played: *'Thank you. After ringing there will be silence while I try to connect you. Any problems and you will be reconnected to an operator.'* Provided the caller does not clear down, the ISAP attempts to connect the call. If there is no reply or the destination number is engaged, the customer is informed and asked to try again later. If it transpires that the destination number is a payphone, the customer is informed that this is not allowed and is offered to be reconnected to an operator. Under normal circumstances, however, the customer will hear ringing tone until the called number answers. On answer, the transmission bridge is switched to prevent communication between the calling and called customers, and the called customer is played the message: *'Good morning/afternoon/ evening. You have a BT automated reverse charge call from....'* — here the original caller's name is played back — '*... from the* <geographical location> *area. If you will pay for the call please say yes after the tone, otherwise say no.. .'* Hearing the caller's name, spoken by themselves, together with the geographical area from which they are calling (this is derived from the caller's telephone number), gives the called customer sufficient information as to whether to accept the call. A simple response of *'yes'*, *'yup'*, or even *'yes please'* causes the ISAP to switch the BTOSS transmission bridge to enable two-way communication. The ISAP then says: *'Go ahead please.'* If the ISAP does not recognise a positive acceptance of the call charges it will ask the called customer: *'If you wish to be connected to an operator please hold the line, otherwise please replace the handset.'* If the called customer clears down, the call is disallowed with the message back to the caller: *'I'm sorry, the number you are calling will not pay for the call. If you require further assistance, please hold for reconnection to an operator, otherwise please replace the handset.'* If either or both customers hold at this point, the call is requeued back to an operator with an appropriate text message to allow them to sort the problem out manually.

5. Results

5.1 Market survey findings

A market survey of trial users suggests that the automation of the services presented here is indeed appropriate, although the level of acceptability of the services trialled does vary. Reverse charge callers and recipients, particularly those over 40 and/or female, appear to have greater difficulty with an automated service because these calls are interactive and customers are deprived of the guidance and reassurance of a human operator. With alarm calls and the advice of duration and charge services, trial users appear to have few concerns about automation since BT is required to play a specific and preplanned role, needing no intervention from the customer. Although those less computer-literate claim that it would still be pleasant to be able to talk to another human being, they acknowledge that in this instance it is not necessary.

5.2 Call-handling time savings

A detailed analysis of the traffic figures for the duration of the trials found that automating:

- referrals and onward connects gave a saving of 10 seconds per call when compared to those calls handled manually,
- alarm call delivery a saving of 38.5 seconds per call,
- advice of duration and charge, 68 seconds,
- reverse charge call, 23 seconds.

These figures also take into account the time taken to deal with calls where a customer has been given an automated service and has subsequently held the line for further assistance. This represents an overall saving of almost 3 seconds per call on the 108 million assistance calls per year.

5.3 Additional comments

A large proportion of reverse charge calls made during the working day are to business premises, often to another operator who is expecting their own people to be contacting the office using the reverse charge service, and will immediately accept the charges. An early trial option allowed for a shorter dialogue when calls were connected to business lines during the peak business period. The message was: *'Good morning/afternoon/evening. You have a BT automated reverse charge call from* <name> *from the* <geographical location> *area. Will you pay for the call?'* Unfortunately this shortened question, without a prompt for *'yes or no'*, received a wide variety of responses including *'Sure, all right then, I will, OK.'* This type of response caused the ISAP yes/no recognition success rate to fall dramatically, and the shortened message was abandoned in favour of the more explicit question designed to elicit a *'yes* or *no'* response.

This sort of experience was reinforced when providing alternative dialogues for other services — a very minor change in the wording, pace or intonation of a message can have a significant effect on the intelligibility of the message. It is essential in a trial system to have the capability of making these sort of changes quickly and the change required very often only becomes obvious when running the service. Pre-trial simulations of the reverse charge dialogue did not indicate any major problems, whereas monitoring the responses during live calls immediately highlighted areas of improvement. These included speeding up the pace of delivery by 20%, extending the length of the beep after asking for the name, and removing the fast-track business option.

One area of concern for the reverse charge service was that customers might abuse the name recording facility in order to pass over a short message to the destination, who could then refuse to accept the call, having got the necessary information for free. To date, no evidence of this occurring has been detected, although the option to disable the record facility exists if it becomes a problem. Market research showed that most customers were happy to accept the call on the basis of geographical location only, without a recorded name.

6. Future opportunities

It is intended that the trial system described in this paper will be rolled out nationally to every BTOSS site. Any future interactive voice service requiring connection or fall-back to an operator can be potentially provided by the BTOSS-ISAP platform combination.

For operator services, the migration of the international operator assistance traffic to a BTOSS-ISAP platform shared with the national operator assistance service offers further automation opportunities for the future.

7. Conclusions

On the basis of the qualitative and quantitative research findings, it appears that the partial automation of operator services is an appropriate step for BT to take nationwide. Although certain customer segments do object quite strongly to the introduction of automation and the consequent diminution of personal communications and relationships, there is also a widespread belief among customers that automation is the way of the future; it is regarded as a fundamental requirement for successful businesses in today's commercial world. In this respect, BT are not alone in what they are doing and are therefore unlikely to be penalised by their customers. Indeed, the more technically sophisticated customers actually perceive the move to automation as a positive step for BT, reflecting a progressive and competitive attitude. Nevertheless, in the light of the seemingly increasing appeal of competitors and in order to appease those customers who are uncomfortable with automation, it is essential that the introduction of BT's automated service is as smooth and sensitive as possible.

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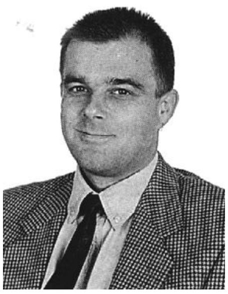

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